



Peritoneum — MCQ Practice

Anatomy GIT Block • MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy • 10 Medium • 6 Hard • suggested time ≈ 25 minutes.

Q Questions

1. Which of the following organs is retroperitoneal?

EASY

- A. Stomach
- B. Pancreas
- C. Spleen
- D. Transverse colon
- E. Jejunum

2. The visceral peritoneum is sensitive to which stimulus?

EASY

- A. Pain from a needle prick
- B. Temperature
- C. Light touch
- D. Stretching and tearing only
- E. Pressure and temperature

3. Which nerve supplies the parietal peritoneum over the central part of the diaphragm?

EASY

- A. Phrenic nerve
- B. Vagus nerve
- C. Intercostal nerves T7–T12
- D. Obturator nerve
- E. Greater splanchnic nerve

4. A peritoneal fold that connects the stomach to another organ is called:

EASY

- A. A mesentery
- B. A mesocolon
- C. An omentum
- D. A ligament
- E. A recess

5. After upper abdominal surgery a patient develops an abscess beneath the diaphragm. Imaging shows two compartments separated by the falciform ligament. Which spaces are these?

MEDIUM

- A. Hepatorenal and subhepatic spaces
- B. Rectouterine and vesicouterine pouches
- C. Supra- and infracolic spaces
- D. Omental bursa compartments
- E. Right and left subphrenic spaces

6. A male patient has peritoneal fluid collecting between the rectum and the urinary bladder. Which pouch is this?

MEDIUM

- A. Rectovesical pouch
- B. Rectouterine pouch
- C. Hepatorenal pouch
- D. Vesicouterine pouch
- E. Subphrenic pouch

7. A patient with a perforated appendix develops pus that tracks alongside the ascending and descending colon. Which structures allow this spread?

MEDIUM

- A. Subphrenic spaces
- B. Hepatorenal pouch
- C. Paracolic gutters
- D. Lesser sac
- E. Sigmoid mesocolon

8. A trauma patient has fluid in the upper peritoneal compartment around the liver, stomach, spleen and superior duodenum. Which compartment is this?

MEDIUM

- A. Infracolonic compartment
- B. Supracolic compartment
- C. Lesser sac
- D. Pelvic cavity
- E. Right paracolic gutter

9. A loop of small intestine herniates from the greater sac into the lesser sac. Through which opening did it pass?

MEDIUM

- A. Transverse mesocolon
- B. Falciform ligament
- C. Hepatogastric ligament
- D. Epiploic (omental) foramen
- E. Inferior duodenal recess

10. A patient lying supine after a liver laceration has free blood in the peritoneum. Where does it collect first?

MEDIUM

- A. Left paracolic gutter
- B. Lesser sac
- C. Left subphrenic space
- D. Infracolonic space
- E. Hepatorenal (Morison's) pouch

11. A surgeon clamps the free margin of the lesser omentum. Which structure lies deepest (most posterior) in it?

MEDIUM

- A. Portal vein
- B. Bile duct
- C. Hepatic artery proper
- D. Right gastric artery
- E. Left gastric vein

12. Why are pelvic peritoneal infections more common in women than in men?

MEDIUM

- A. The female peritoneum is thinner
- B. Women lack a greater omentum
- C. The female peritoneal cavity communicates with the exterior via the uterine tubes, uterus and vagina
- D. The female pelvis has no paracolic gutters
- E. The obturator nerve does not supply the female peritoneum

13. At laparotomy the greater omentum is found wrapped around an inflamed appendix. Which feature allows it to fight the infection?

MEDIUM

- A. Its gastroepiploic arteries
- B. Its fat content
- C. Its attachment to the transverse colon
- D. Aggregations of macrophages (milky spots)
- E. Its phrenic nerve supply

14. Which fold divides the peritoneal cavity into supracolic and infracolic compartments and carries the middle colic vessels?

MEDIUM

- A. Mesentery of the small intestine
- B. Transverse mesocolon
- C. Sigmoid mesocolon
- D. Lesser omentum
- E. Falciform ligament

15. During surgery the surgeon passes a finger through the epiploic foramen and presses backwards. Which structure forms the posterior boundary he feels?

HARD

- A. Portal vein
- B. Caudate lobe of the liver
- C. First part of the duodenum
- D. Head of the pancreas
- E. Inferior vena cava

16. During a sigmoid colectomy, the surgeon dissects the apex of the inverted-V root of the sigmoid mesocolon. Which structure lies here and is at risk?

HARD

- A. Left ureter
- B. Right ureter
- C. Inferior mesenteric vein
- D. Left gonadal vein
- E. Left obturator nerve

17. A pelvic abscess irritates the parietal peritoneum on the peripheral pelvic wall. Which nerve carries this sensation?

HARD

- A. Phrenic nerve
- B. Pelvic splanchnic nerves
- C. Pudendal nerve
- D. Obturator nerve
- E. Subcostal nerve

18. A posterior gastric ulcer perforates into the lesser sac. Which structure forms part of the posterior wall of the space it enters?

HARD

- A. Lesser omentum
- B. Caudate lobe
- C. Pancreas
- D. Gastrosplenic ligament
- E. Transverse mesocolon

19. After cholecystectomy, pus from the hepatorenal pouch reaches the rectouterine pouch. Which pathway connects them?

HARD

- A. Left paracolic gutter
- B. Epiploic foramen
- C. Lesser sac
- D. Root of the mesentery
- E. Right paracolic gutter

20. The root of the small-intestine mesentery runs from the left side of L2 to the right sacroiliac joint. Which structure does it cross?

HARD

- A.** Left ureter
- B.** Right ureter
- C.** Left kidney
- D.** Sigmoid colon
- E.** Inferior mesenteric vein

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	The slides list kidneys, suprarenals, ureters, pancreas, duodenum, ascending and descending colon as retroperitoneal. The others listed are intraperitoneal.
2	D	EASY	Visceral peritoneum is innervated by autonomic afferents and is sensitive only to stretching and tearing; parietal peritoneum senses pain, touch, pressure and temperature.
3	A	EASY	The central diaphragmatic peritoneum is supplied by the phrenic nerve; the peripheral part by T7-T12 and L1.
4	C	EASY	Omenta connect the stomach with other organs; mesenteries connect intestine to the posterior wall; ligaments connect organs to the wall or each other.
5	E	MEDIUM	The subphrenic space, below the diaphragm and above the liver, is divided into right and left by the falciform ligament.
6	A	MEDIUM	In males the rectovesical pouch lies between rectum and bladder.
7	C	MEDIUM	The paracolic gutters run beside the ascending and descending colon; the right one connects the hepatorenal pouch with the pelvis.
8	B	MEDIUM	The supracolic compartment lies above the transverse colon and its mesocolon and contains the liver, stomach, spleen and superior duodenum.
9	D	MEDIUM	The greater and lesser sacs communicate only through the epiploic foramen.
10	E	MEDIUM	The hepatorenal pouch is the lowest recess of the abdomen in the supine position, so blood and fluid collect there first.
11	A	MEDIUM	In the hepatoduodenal ligament the bile duct is anterior-right, the hepatic artery proper anterior-left and the portal vein posterior and deepest.
12	C	MEDIUM	In males the peritoneum is a closed sac; in females it opens to the exterior through the ostia of the uterine tubes, uterus and vagina.
13	D	MEDIUM	Milky spots are macrophage patches; the greater omentum moves to infection and seals it off, earning the name 'policeman of the abdomen'.
14	B	MEDIUM	The transverse mesocolon, fused with the back of the greater omentum, separates the compartments and contains the middle colic vessels.
15	E	HARD	Epiploic foramen: anterior hepatoduodenal ligament (portal triad), posterior IVC, superior caudate lobe, inferior first part of the duodenum.
16	A	HARD	The apex of the sigmoid mesocolon lies where the left ureter crosses the left common iliac vessels.
17	D	HARD	The peripheral part of the pelvic peritoneum is supplied by the obturator nerve.
18	C	HARD	The lesser sac lies behind the stomach; its posterior wall is formed by the pancreas, left kidney, left adrenal, aorta and IVC.
19	E	HARD	The right paracolic gutter connects the hepatorenal pouch with the pelvis, a route for spread of infection from abdomen to pelvis.
20	B	HARD	From above down it crosses the 3rd part of the duodenum, aorta, IVC, right gonadal vessels, right ureter and right psoas major.